

# IIMT COLLEGE OF ENGINEERING

Knowledge Park III , Greater Noida

(Department of Computer Science and Engineering-AI)

## **AI-BASED STUDENT DROPOUT PREDICTION & COUNSELING SYSTEM**

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# INTRODUCTION

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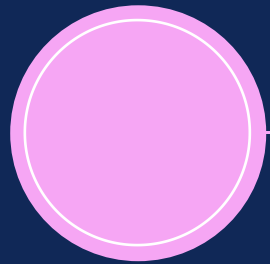
The idea of the project is to predict the dropout risk of students using academic, personal, and behavioral factors and provide proper counseling recommendations to reduce their dropout possibility. The system analyzes student performance data and predicts whether the student is at **low, medium, or high risk** and then provides **personalized guidance or counseling suggestions**.

# OBJECTIVES

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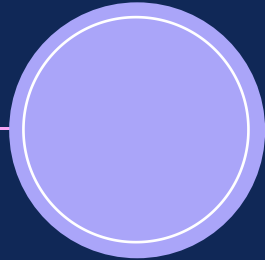
- Predict student dropout risk using academic and behavioral data.
- Categorize students as Low, Medium, or High risk.
- Provide personalized counseling recommendations.
- Help institutions reduce dropout rates.

# PLANNING OF WORK



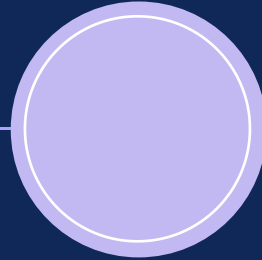
## Step 1 ●

Problem  
Identification



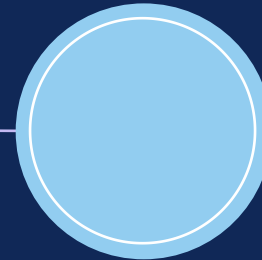
## Step 2 ●

Data Collection  
& Feature  
Identification



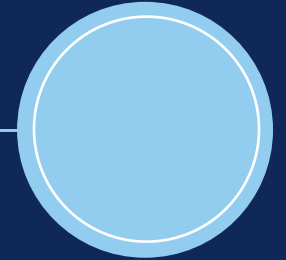
## Step 3 ●

Data  
Normalization



## Step 4 ●

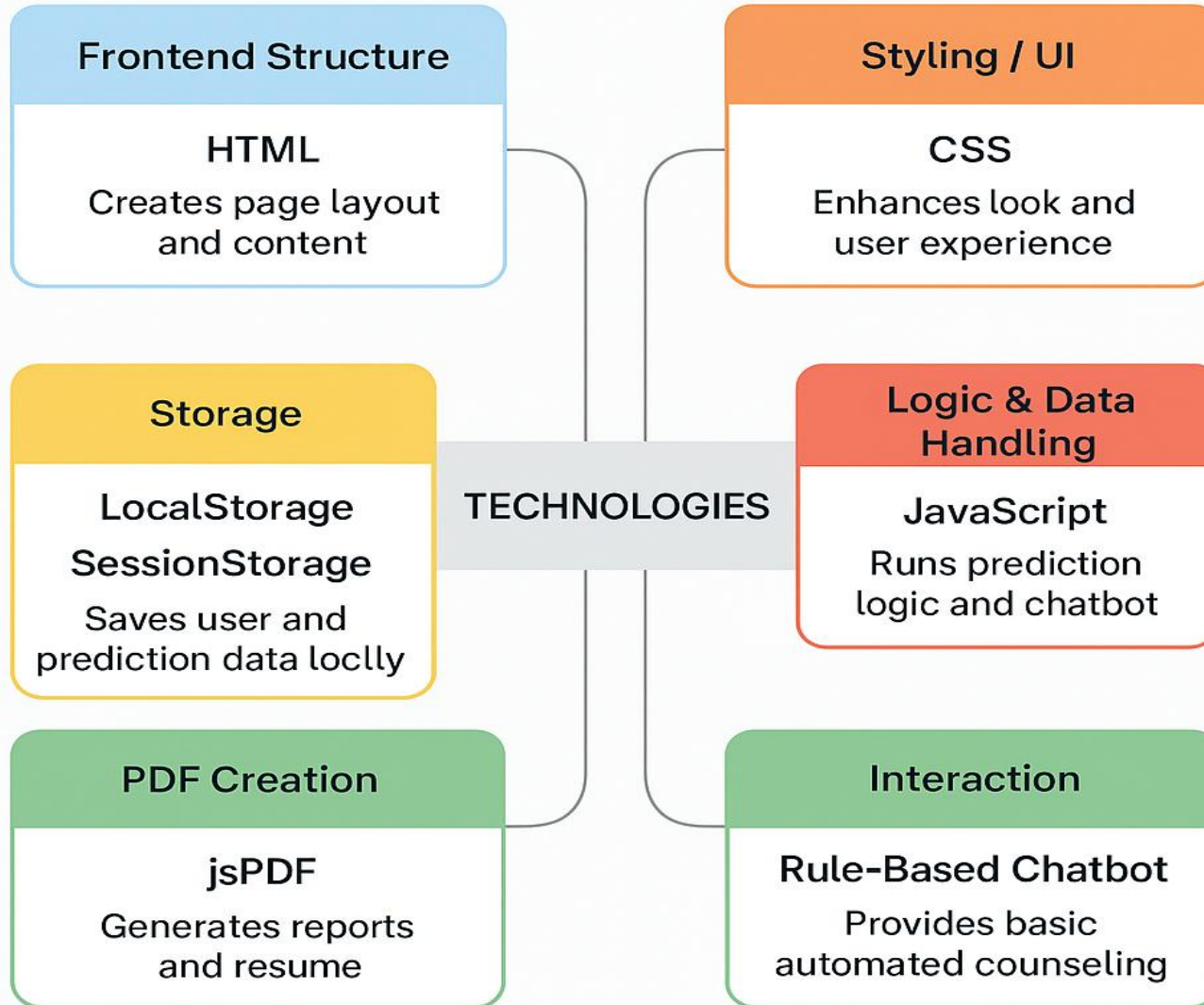
Risk Prediction  
Model



## Step 5 ●

Counselling  
Suggestion  
Generation and  
System  
Implementation

# METHODOLOGY



# System Features

- Dropout Risk Prediction Module
- Personalized Counseling Suggestions
- Chat bot for quick student support
- Daily Routine Tracker
- To-Do Task Manager
- Resume Builder

# IMT COLLEGE OF ENGINEERING, GREATER NOIDA

Department of Information Technology

## Department Production

2022-23

2022-23

2022-23

2022-23

2022-23

2022-23

2022-23

2022-23

2022-23

2022-23

2022-23

2022-23

### Sign in to Dashboard

Create an account or sign in. Data will remain local to your browser.

Username

Password

Login

Sign Up

Continue as Guest

Snapshot of Login Page

Student Minimal  
Dashboard

A compact dashboard for student  
support tools

Home

Dropout Prediction

Student Counselling

Daily Routine Tracker

To-Do Maker

Resume Builder

Feedback

Data stored locally in browser  
(localStorage)

Reset All Data

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A minimal dashboard to predict student dropout risk and provide counseling by developing an AI-based system.

Guest • v1.0

Logout

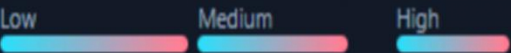
Dark

Home — Overview

Dropout predictions (last 30)

0

Low / Medium / High risk



Quick actions

New Prediction

Build Resume

Project details

This minimal dashboard demonstrates a client-side toolkit for student dropout prediction, counselling, routine tracking, to-do management, resume generation, and feedback capture. All data is stored locally (no server).

Built for demo • All data local • Open-source minimal UI

# SNAPSHOT OF WORK

## Login Page

### Sign in to Dashboard

Create an account or sign in. Data will remain local to your browser.

Username

username

Password

password

Login

Sign Up

Continue as Guest

Enter username and password

## Signup Page

### Sign in to Dashboard

Create an account or sign in. Data will remain local to your browser.

Full Name

Full Name

Email

email@example.com

Username

username

Password

password

CAPTHA (4-digit simulated)

####

Enter Captha

Verify & Sign Up

Continue as Guest

Enter username and password

# Prediction system

## IIMT COLLEGE OF ENGINEERING, GREATER NOIDA

Logout

Dark

A minimal dashboard to predict student dropout risk and provide counseling by developing an AI-based system.

### Dropout Prediction

Student name

Full name

Student ID

ID or roll no

Attendance %

85

GPA (0-10)

7.2

Assignments completed %

75

Engagement score (0-100)

60

Parental education

Postgraduate or above



Financial stress (0-100)

30

Past backlogs / arrears (count)

0

Health / Stress indicator (0-100)

20

Extra-curricular engagement (0-100)

40

Socioeconomic risk

Low



Predict

Download Report

No prediction yet

History

Export CSV

Clear

# Counseling system

## Student Minimal Dashboard

A compact dashboard for student support tools

Home

Dropout Prediction

Student Counselling

Daily Routine Tracker

To-Do Maker

Resume Builder

Feedback

## IIMT COLLEGE OF ENGINEERING, GREATER NOIDA

Guest • v1.0

Logout

Dark

A minimal dashboard to predict student dropout risk and provide counseling by developing an AI-based system.

### Student Counselling

Student

Type

Name or ID

Academic

Notes / Session summary

Counsellor notes and suggested actions

Save Session

Export Sessions

Saved sessions

### Counselling Chatbot

quick replies • local only

Hi — I am the counselling assistant. Ask about attendance, GPA, assignments, study plans, stress, or careers. The chat is stored locally only.

Ask about attendance, study plan, stress, career...

Send

Save Chat

Clear

Bot uses simple heuristics to provide suggestions

## Daily Routine Tracker

### Daily Routine Tracker

Activity

e.g. Study Data Structures

Add

Export

Clear

## To-Do List Maker

### To-Do List Maker

Add a task

Download .txt

Clear

## FUTURE SCOPE

- Integrate real-time college database.
- Use advanced ML models for improved accuracy.
- Expand counseling to mental health and career planning.
- Develop mobile app version.

# REFERENCES

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- [www.stackoverflow.com](http://www.stackoverflow.com)
- [www.wikipedia.com](http://www.wikipedia.com)
- <https://www.geeksforgeeks.org/>
- <https://www.javatpoint.com/>
- <https://www.w3schools.com/>
- [www.youtube.com](http://www.youtube.com)



**THANK YOU**

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